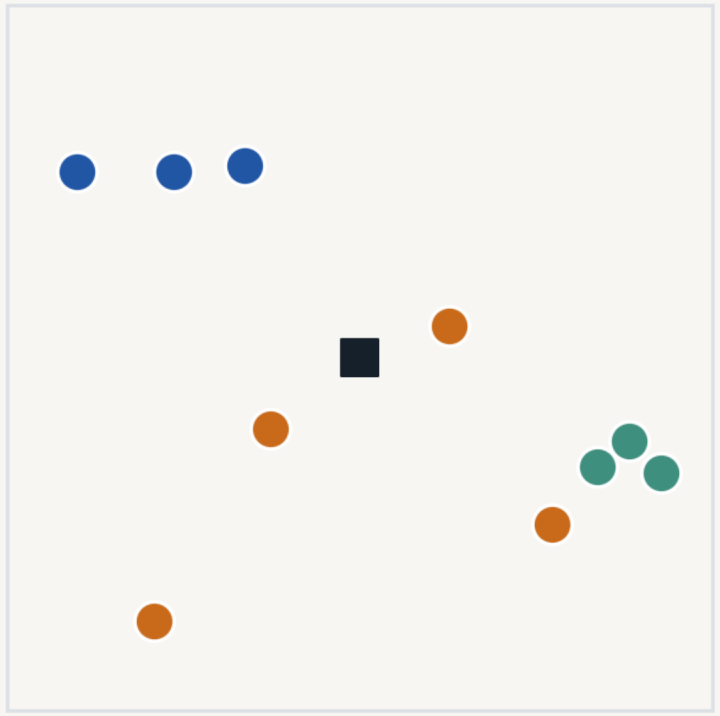
PASQAL CHALLENGE · TRACK A · VEHICLE ROUTING

Vehicle routing on Pasqal atoms: one atom per customer, placed like the map

4 / 10 / 15 atoms instead of 14 / 45 / 80

Byteskip · Atom Wang, Lucas Ma

The problem: customers into trucks, fewest atoms



tier	customers	trucks	capacity
S	4	2	7
M	10	3	20
L	15	4	30

Two models

	Model A · direct QUBO	Model B · seeds
a bit means	customer i is on truck k	customer i is a truck’s center
atoms S / M / L	4 / 30 / 60	4 / 10 / 15
atom layout	M, L: impossible	the customer map
result	S only	all 9 instances

A Idea: write the cost directly, no slack bits

$$\text{Total Loss} = C_{\text{cluster}} + \lambda P_{\text{capacity}}$$

$$C_{\text{cluster}} = \sum_{\text{pairs}} d_{ij}\,(1 - x_i - x_j + 2x_i x_j) \quad \rightarrow \textbf{optimise}$$

$$P_{\text{capacity}} = (3x_A + 4x_B + 2x_C + 3x_D - 6)^2 \quad \rightarrow \textbf{pass the line}$$

A QUBO → machine energy

our QUBO

$$\text{Total Loss} = \text{const} + \sum_i a_i\, x_i + \sum_{i<j} b_{ij}\, x_i x_j$$

Pasqal energy

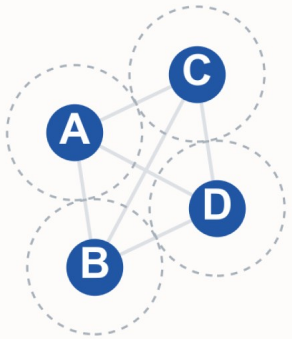
$$E = \sum_i (-\delta)\, x_i + \sum_{i<j} \frac{C_6}{r_{ij}^6}\, x_i x_j$$

term	match?
per bit	no: a differs, −δ is one value
per pair	approx.: b set by distance r
pair sign	yes: both positive

A Atom layout

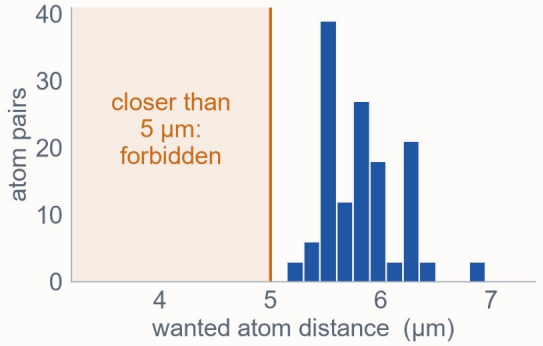
$$r = \left(\frac{C_6}{\text{wanted coupling}} \right)^{1/6}$$

r ≥ 5 µm · radius ≤ 38 µm



S · 4 atoms

fits, error η = 0.54–0.62



M · 30 atoms

does not fit

A Laser and result: S only

instance	usable answers	best answer	random 0/1, best
S · seed 0	5 %	3 %	13 %
S · seed 1	90 %	4 %	15 %
S · seed 2	96 %	95 %	15 %
M, L	no layout	—	—

Why A fails, and what B changes

	far customers	near customers
model A wants	strong coupling	weak coupling
atoms give	weak (far atoms)	strong (near atoms)
model B wants	weak: both can be centers	strong: not both centers

B Idea: pick a center for each truck



a bit means	customer i is a truck’s center (★)
atoms	one per customer: 4 / 10 / 15
the machine	picks K centers far apart
then	every other customer joins the nearest center with room

B QUBO = machine energy

$$E = -\delta \sum_i x_i + \sum_{i<j} \frac{C_6}{(\alpha\, d_{ij})^6}\, x_i x_j$$

term	model B wants	Pasqal gives	match?
per bit	the same −δ for all	−δ, the same for all	yes
per pair	$C_6 / (\alpha\ d)^6$	C_6 / r^6 with r = α d	yes, by layout
capacity	not in the energy	—	checked after

B Atom layout: the map, scaled

$$r_{ij} = \alpha\, d_{ij}, \qquad \alpha = \frac{R_b}{r_c}, \qquad R_b = \left(\frac{C_6}{\Omega}\right)^{1/6}$$

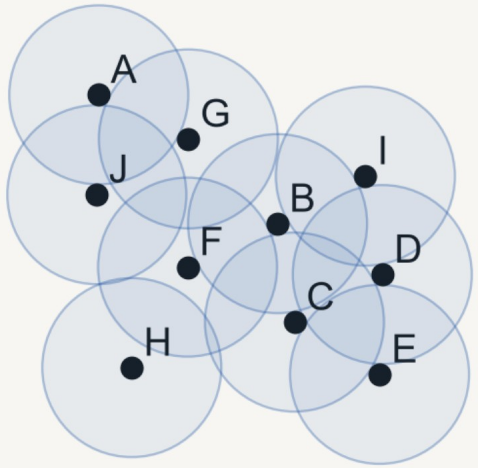
	S	M	L
atoms	4	10	15
blockade radius	7 µm	9 µm	9 µm
layout radius (≤ 38)	6–8 µm	10–11 µm	14 µm
Pulser device check	pass	pass	pass

B Laser: three knobs

knob	tried	per	chosen S / M / L
strength Ω	blockade radius 7, 9, 12 μm	tier	7.36 / 1.63 / 1.63 rad/μs
duration T	2000, 4000, 6000 ns	tier	4000 / 2000 / 6000 ns
final detuning δ	c = 0.7 … 1.9 × window middle	instance	c = 1.3–1.9
picked by	share of best answers, then usable	—	1000 shots each

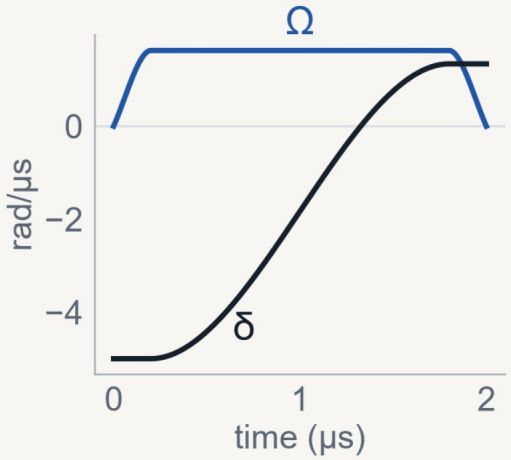
B How it runs, M_seed0

1 · atoms on the map



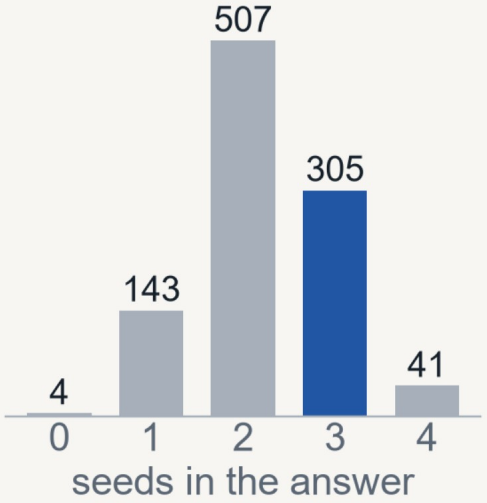
circles overlap → cannot both be seeds

2 · laser, 2 µs



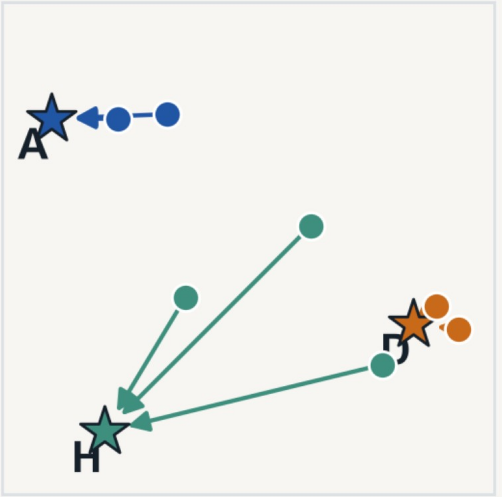
Ω on, δ from −5.0 to 1.34

3 · read 1000 times



305 answers have exactly 3 seeds

4 · join nearest seed with room



loads 20 / 19 / 18 ≤ 20 · **= the optimum**

B Result: share of answers that are the best

tier	machine	random 0/1	random K seeds
S	51–91 %	25–27 % (2–4×)	67 % (0.8–1.4×)
M	13–20 %	1.6–2.9 % (7–12×)	13–30 % (0.7–1.1×)
L	2–32 %	0–0.2 % (≥ 60×)	2.8–4.0 % (0.7–8×)

Summary

NEW

Capacity with no slack bits: 4 / 30 / 60 atoms

One atom per customer: **4 / 10 / 15 atoms**

Best answer more often than random 0/1: 2–4× S, 7–12× M, ≥ 60× L

DID NOT WORK

Direct QUBO on M, L: no 2D layout

About half of M, L answers still unusable

Balance penalty costs up to 13.8 % before any shot